

Minh Quyen Nguyen

517-977-2108 | nguy1310@msu.edu | github.com/iUV-1 | <https://www.linkedin.com/in/mquyennguyen/> | E. Lansing, MI

EDUCATION

Michigan State University

E. Lansing, MI

Bachelor of Science in Computer Science

May 2027

- Minor in Game Design and Development
- Concentration in Multimedia and Graphics
- GPA: 3.6

TECHNICAL SKILLS

Languages: Python, C# .NET, C++, JavaScript/TypeScript, SQL

Frameworks: Unity, ReactJS, NextJS, WPF, Prisma ORM

Developer Tools: Git, GitHub, Docker, Unix, PowerShell, SCCM, Active Directory, Network Config

EXPERIENCE

MSU College of Engineering Division of Engineering Computing Service

E. Lansing, MI

Student Info Tech Assistant I

Feb. 2024 – Present

- Administered a fleet of 500+ Windows lab computers using Microsoft SCCM and Active Directory (AD) to manage centralized user access, system configurations, and security policy enforcement.
- Engineered and executed automated, large-scale software deployments to hundreds of machines, ensuring consistent access to high-performance engineering applications for 5,000+ students and faculty.
- Streamlined routine system maintenance by developing custom PowerShell scripts for remote troubleshooting and health monitoring, significantly reducing manual overhead and maximizing lab uptime.
- Developed internal management tools and desktop applications using C# and WPF, replacing manual CLI-based workflows with intuitive XAML-based graphical interfaces to consolidate diagnostic data.

Nghia Nippers

Ho Chi Minh City, Vietnam

Software Engineer Intern

May 2025 – Aug. 2025

- Developed a full-stack web application using Next.js, architecting a responsive UI that ensures a seamless user experience across mobile, tablet, and desktop devices.
- Engineered a secure authentication system leveraging password encryption and JSON Web Tokens (JWT) for stateless session management, integrated with a Next.js frontend.
- Implemented Role-Based Access Control (RBAC) to enforce granular permissions, dynamically rendering UI components and restricting API access based on verified user roles.
- Designed and optimized database schemas using Prisma ORM with PostgreSQL, ensuring efficient data persistence and high-performance queries for user-generated content.
- Containerized the deployment workflow using Docker, creating reproducible images that streamlined the development-to-production pipeline.

PROJECTS

Parallel Ray Tracing & Physics Engine | C, Parallel Computing

- Achieved a 20x performance speedup by transforming a baseline scalar engine into a highly optimized parallel system using SIMD intrinsics and Cilk/Multi-threading.
- Engineered a Bounding Volume Hierarchy (BVH) to replace brute-force intersection tests, reducing algorithmic complexity from $O(n)$ to $O(\log n)$.
- Implemented advanced spatial pruning and physics culling to reduce redundant computations.

Real-Time Software Rasterizer | C++, SDL3, Dear ImGui, Linux - github.com/iUV-1/SoftwareRenderer

- Developed a multi-stage 3D rendering pipeline from scratch on the CPU, implementing custom vertex processing, homogeneous clipping, and perspective-correct texture mapping.
- Achieved a 3x speedup over the naive implementation by optimizing the fragment stage with tile-based rendering and SIMD vectorization to process pixel blocks in parallel.
- Integrated SDL3 for low-level frame-buffer access and Dear ImGui to build a real-time telemetry dashboard for monitoring frame times and adjusting shader parameters on the fly.
- Optimized memory bandwidth and cache locality through scanline conversion algorithms and a linear depth buffer, maintaining stable performance in a Linux environment.